

Data Flow Diagram

Date	12 July 2026
Team ID	
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	2 Marks

Data Flow Diagram

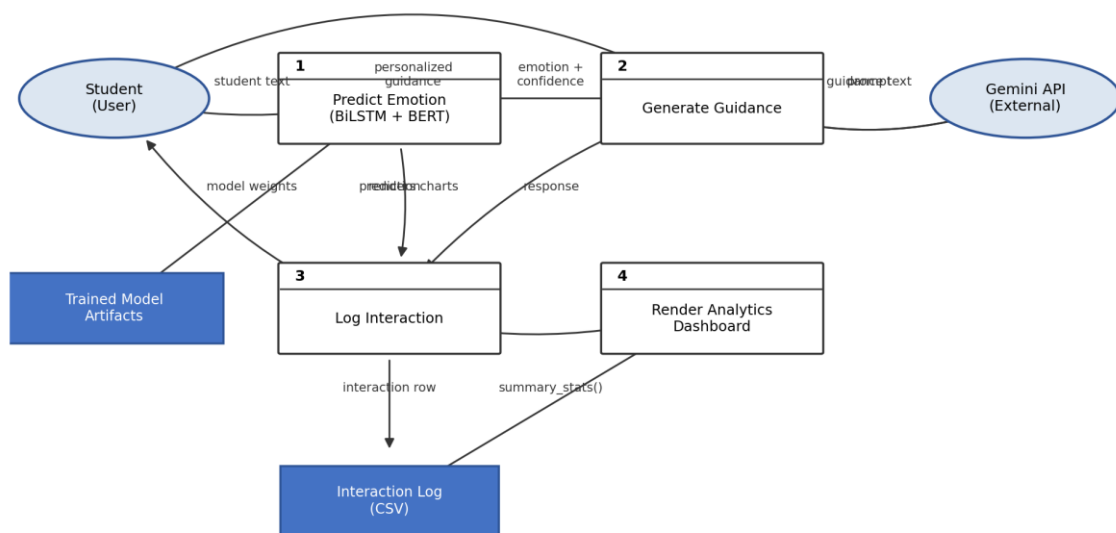
Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person or system outside the project that sends or receives data (Student, Gemini API).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g., Predict Emotion, Generate Guidance).
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (Interaction Log CSV, Trained Model Artifacts).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

AI Learning Assistant — Data Flow Diagram

Data Flow Diagram — AI Learning Assistant



Description: The Student (external entity) submits free-text input, which Process 1 (Predict Emotion) classifies using BiLSTM and BERT models loaded from the Trained Model Artifacts data store. The predicted emotion and confidence flow to Process 2 (Generate Guidance), which exchanges a prompt and response with the Gemini API (external entity) and returns personalized guidance to the student. Both the prediction and the response are passed to Process 3 (Log Interaction), which writes a row to the Interaction Log data store. Process

4 (Render Analytics Dashboard) reads summary statistics back from that data store and renders charts back to the student.